

InuaBiz

Product Requirements Document (PRD) — v1.1

Multi-Tenant Micro-POS & Business Operations Engine for Kenyan Small-Scale Vendors

Project Lead:	Zachariah Komu Nyokabi (Full-Stack / AI Engineer)	Target Market:	Nairobi & Regional Kenyan MSMEs
Tech Stack:	React, Tailwind, Supabase, Edge Functions	Date:	August 2026

1. Executive Summary & Vision

InuaBiz ("Lift Your Business") is a lightweight, mobile-first SaaS platform engineered specifically for micro, small, and medium-scale enterprises (MSMEs) in Kenya—including retail dukas, boutiques, chemists, and neighborhood eateries. High setup friction, bloated software design, and complex registration hurdles prevent small vendors from adopting standard POS systems. InuaBiz resolves this by offering zero-friction onboarding, instant localized M-Pesa integration, discrete customer loyalty tracking, and AI-driven predictive cash-flow insights.

Core Mission

To empower Kenyan small vendors with modern digital tools to digitize daily inventory, track customer credit (Duka Debt/Mani), streamline payments, and gain automated financial intelligence—without requiring heavy hardware or formal corporate structures.

2. Target Audience & Personas

Persona Category	Target User Profile	Primary Pain Points	InuaBiz Solution
Micro-Vendor / Duka	Mama Njoroge (Retail Grocery Owner)	Manual ledger books, lost credit records, unverified M-Pesa text messages.	2-tap credit recording, instant M-Pesa notification parsing, mobile-first UI.
Specialty Retail	Boutique & Chemist Operators	Stock leakage, unknown profit margins, unpredictable reorder cycles.	Stock alerts, AI demand forecasting, low-stock reorder triggers.
Super-Admin	Solo System Developer / Operator	Managing multiple vendors manually, debugging user issues, operational visibility.	GIS Store Map, central tenant control, unclaimed payment queues, one-click impersonation.

3. Low-Friction Onboarding Flow (< 2 Minutes)

The onboarding journey is designed to minimize drop-off rates on mobile devices by postponing complex business registration requirements and letting vendors immediately start recording transactions.

- 1. **Phone-First Authentication (30s):** Vendor enters Kenyan mobile number (07xx / 01xx). Supabase Auth issues a 4-digit SMS OTP code for single-tap login.
- 2. **Business Details & GPS Pinning (45s):** Input Business Name and select Category (Duka, Boutique, Chemist, Hardware). One-click browser geolocation ("Detect My Location") tags the store coordinates for GIS mapping.
- 3. **Payment Destination Setup (30s):** Vendor inputs their active receiving mechanism (Personal M-Pesa Number, Till Number, or Paybill). A 14-day full trial launches instantly.
- 4. **Guided Micro-Tour (15s):** User lands directly on the POS screen with a pre-loaded sample product to complete their first test checkout.

4. Core Features & Functional Requirements

4.1 VENDOR MOBILE-FIRST APPLICATION

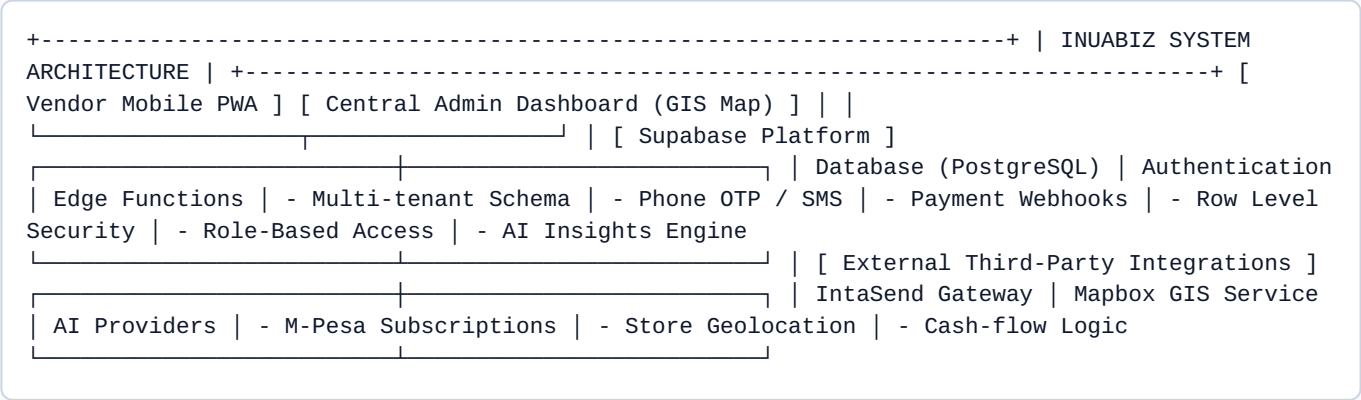
Feature Module	Description & Key Functionality	Tech Implementation
Mobile POS & Checkout	Fast grid search, barcode scanning via mobile camera, instant cart total calculation, and custom discount toggles.	React, Web Barcode Detector API, Tailwind UI.
M-Pesa Multi-Channel Sync	Parses payment confirmations across Paybill, Till, Pochi la Biashara, or personal M-Pesa numbers with auto-reconciliation.	Supabase Edge Functions, IntaSend Webhooks / M-Pesa SMS Reader.
Duka Debt / Credit Ledger	Digital book for customer credit ("Kukopesha"). Tracks outstanding balances, sends automated WhatsApp payment reminders.	Supabase RLS, WhatsApp Deep-Linking / Twilio API.
Discrete Loyalty Tracking	Quietly tracks frequent buyers via phone numbers without needing physical loyalty cards or app downloads for customers.	PostgreSQL Aggregations, Customer Profiles.
Inventory & Stock Alerts	Real-time inventory levels, low-stock push notifications, product cost vs. selling price margin tracking.	PostgreSQL Triggers & Realtime Channels.
AI Financial Intelligence	Provides predictive cash-flow forecasting, best-selling product predictions, and inventory reorder recommendations.	OpenAI / Claude API integrated via Supabase Edge Functions.

4.2 CENTRAL SUPER-ADMIN COMMAND CENTER

Designed specifically for the solo platform owner to maintain complete operational observability and control over all onboarded tenants.

Admin Capability	Operational Focus
Live GIS Vendor Map	Interactive Mapbox map showing store locations. Color-coded markers highlight real-time status (● Active, ● Inactive, ● Error/Failed Webhook, ● Suspended). Includes regional density heatmaps.
One-Click Impersonation	Shadow session creation allowing the admin to jump directly into any vendor's view for instant support and live troubleshooting without asking for credentials.
Unclaimed Payment Queue	Reconciliation dashboard for inbound payment webhooks that failed automated tenant matching, allowing manual assignment to the correct vendor.
Subscription & MRR Analytics	Tracks Monthly Recurring Revenue, active trials, conversion rates, and automated billing retries via IntaSend.
Platform Health & Broadcasts	Monitors DB connection health, Edge Function latency, AI API consumption costs, and provides system-wide broadcast banner messaging.

5. System Architecture & Tech Stack



6. Pricing & Subscription Model

- **Free Trial:** 14 days full access (no credit card or upfront business paybill required).
- **Flat Subscription (KES 3,000 / month):** All-inclusive access to the entire InuaBiz platform—Mobile POS, M-Pesa auto-reconciliation, credit ledger, GIS mapping, multi-device access, and full AI Financial Intelligence insights.
- **Payment Gateway:** Subscriptions collected via IntaSend to bypass lengthy Safaricom business registration processes in early phases.
- **Tier Flexibility:** Single flat rate structure for immediate launch simplicity; tier splitting can be introduced in later iterations if necessary.